Electric Machines Course Project

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Objective

To design a 1 phase power transformer for a solar panel.

Design Case Statement:

- 1 phase power transformer
- Up to 35 kVA
- Secondary side voltage is 14.4 kV
- Primary side voltage is 0.660 kV
- Operating frequency is 60 Hz

Assumptions

- The magnetic flux is uniform over the cross-section of the core
- The material of the core is homogeneous
- The cross-sectional area of the core will not change
- Flux leakage can be ignored

Calculations

The primary side conductor must support the power to be delivered to the secondary side. This power was set to 14.4 kV in the design case statement. The secondary side current can be calculated using equation (1), Where $|\boldsymbol{I}_s|$, $|\boldsymbol{S}_s|$, and $|\boldsymbol{V}_s|$ are the magnitudes of the secondary side's current, apparent power, and voltage, respectively.

$$|\boldsymbol{I}_s| = \frac{|\boldsymbol{S}_s|}{|\boldsymbol{V}_s|} \tag{1}$$

$$|\boldsymbol{I}_s| = \frac{35 * 10^3}{14.4 * 10^3} = 2.431\ A \tag{2}$$

The AWG table has AWG 22, with a rated current density of 214 circular mil / A and an ampacity of 3 A, which will be the wire for the secondary side. Equation (3) shows the process of calculating primary side current, $|\boldsymbol{I}_P|$, where a_V is equal to the transformer voltage ratio.

$$|\boldsymbol{I}_P| = \frac{|\boldsymbol{I}_s|}{a_V} \tag{3}$$

$$a_V = \frac{|\boldsymbol{V}_P|}{|\boldsymbol{V}_s|} \tag{4}$$

$$a_V = \frac{660}{14.4 * 10^3} = 0.04583 \tag{5}$$

$$|\boldsymbol{I}_P| = \frac{2.431}{0.04583} = 53.03 \tag{6}$$

Thus, the primary side windings are set for AWG 4, which has an ampacity of 70 A and a current density of 596 circular mil / A.

A transformer core is selected by its flux density. Using magnetic material tables, the material for the core was selected to be Magnetic material Corporation's Square50, having a flux density of 15K Gauss for the core.

The product of window area, W_C and cross-sectional area, A_C is seen in equation 7, with P (power) in watts, s (current density) in circular mils/A, f (frequency) in Hz, and B (flux density in Gauss). The core type is set to be a core-type (single window). Figure 1 shows a diagram of the core, with calculated measurements included.

$$A_C W_C = \frac{17.26 * P * s}{f * B} \tag{7}$$

$$A_C * W_C = \frac{17.26 * 35 * 10^3 * 596}{60 * 15 * 10^3} = 400\ inch^4 \tag{8}$$

Several design recommendations set $W_C = (7 - 9) * A_C$.

$$Let\ A_C = 7\ inch^2 \tag{9}$$

$$W_C = \frac{A_C * W_C}{A_C} \tag{10}$$

$$W_C = \frac{400}{7} = 57.14\ inch^4 \tag{11}$$

Let the length of the window be twice the width.

$$L_W = 2 * W_W \tag{12}$$

$$W_C = L_W * W_W = 2 * W_W * W_W \tag{13}$$

$$W_W = \sqrt{\frac{W_C}{2}} \tag{14}$$

$$W_W = \sqrt{\frac{57.14}{2}} = 5.36\ inch \tag{15}$$

$$L_W = 2 * 5.36 = 10.72\ inch \tag{16}$$

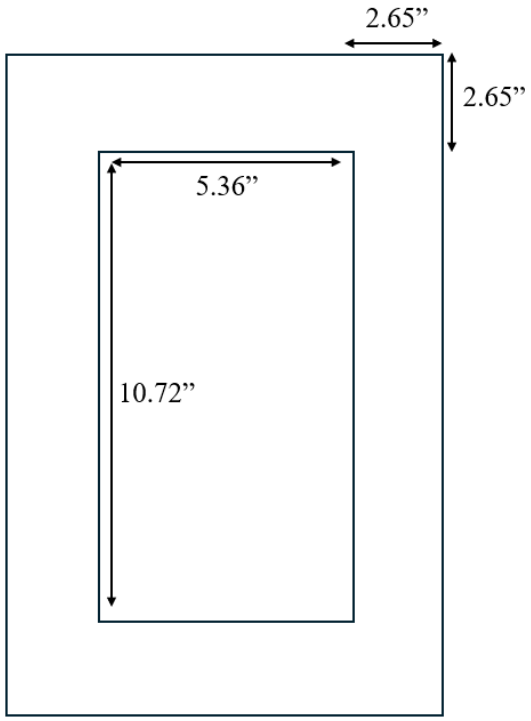


Figure 1. Diagram of magnetic core and its dimensions in inches. Not to scale.

For the power transformer, the turns/volt index (N/V) is found.

$$\frac{N}{V} = \frac{3.5 * 10^6}{f * A_C * B} \tag{17}$$

$$\frac{N}{V} = \frac{3.5 * 10^6}{60 * 7 * 15 * 10^3} = \frac{5}{9} = 0.56 \tag{18}$$

The number of turns on the primary side, N_P, is found with.

$$N_P = |\boldsymbol{V}_P| * \frac{N}{V} \tag{19}$$

$$N_P = 660 * \frac{5}{9} = 367\ turns \tag{20}$$

Using an assumed efficiency, n as 85% The value of Ns is calculated as:

$$N_S = \frac{|\boldsymbol{V}_S| * N_P}{|\boldsymbol{V}_P| * \sqrt{n}} \tag{21}$$

$$N_S = \frac{14.4 * 10^3 * 367}{660 * \sqrt{.85}} = 8686\ turns \tag{22}$$

Therefore, Ns (compensated) is Y turns, and Ns (uncompensated) can be seen below.

$$N_S\ (uncompensated) = \frac{N}{V} * |\boldsymbol{V}_S| \tag{23}$$

$$N_S\ (uncompensated) = \frac{5}{9} * 14.4 * 10^3 = 8000 \tag{24}$$

For a design check, the mitigation of $\boldsymbol{V}_S$ can be quantified using VR%.

$$VR = \frac{|\boldsymbol{V}_S|_{compensated} - |\boldsymbol{V}_S|_{uncompensated}}{|\boldsymbol{V}_S|_{uncompensated}} \tag{25}$$

$$VR = \frac{(N_S)_{compensated} - (N_S)_{uncompensated}}{\frac{N}{V}} \tag{26}$$

$$VR = \frac{8686 - 8000}{\frac{5}{9}} = 1234.8 \tag{27}$$

$$VR\% = \frac{VR}{|\boldsymbol{V}_S|} \tag{28}$$

$$VR\% = \frac{1234.8}{14.4 * 10^3} = 0.086\% \tag{29}$$

Design Outcome

Quantity	Design
Core	Material: Square50 Type: Core-type $A_C = 2.65 * 2.65 = 7\ in^2$ Window: 10.72* 5.36 in^2
Primary Side	Conductor: AWG 4 $N_P = 367\ turns$
Secondary Side	Conductor: AWG 22 $N_S = 8686\ turns$

Remarks

The solar panel must have a rectifier to convert it to an AC signal. This transformer does not include the rectifier. The transformer is designed to be operated below 60 degrees Celsius.